Affine Polynomial Maps and Coordinate-Ring Duality

An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1] *Mindverse Computing LLC, Lynnwood, WA 98087, USA*

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Abstract

The images of the coordinate functions determine the point map. Relations in the source coordinate ring guarantee that the reconstructed point lies in the intended target. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AG-05 / Algebraic Geometry / Report v1

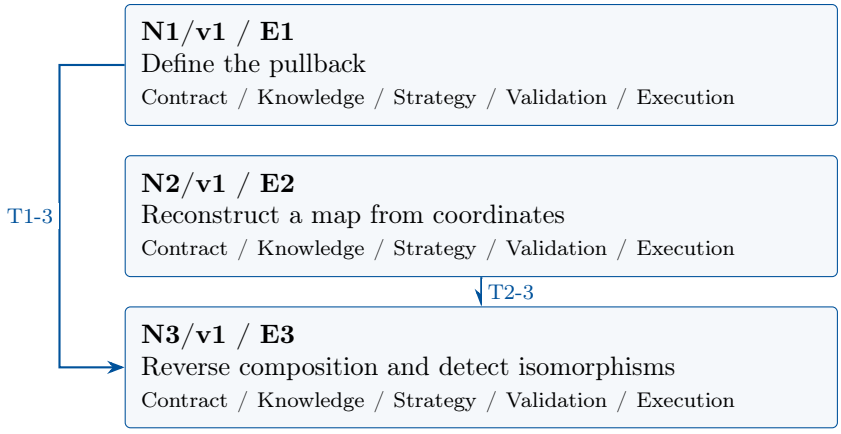
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For affine algebraic sets $X \subset k^n$ and $Y \subset k^m$, polynomial maps $F : X \to Y$ are in bijection with unital k-algebra homomorphisms $\phi : k[Y] \to k[X]$ via pullback $F^*(h) = h \circ F$. Composition is reversed. A polynomial map is an isomorphism with polynomial inverse if and only if its pullback is a k-algebra isomorphism. Here polynomial maps mean maps whose coordinate functions are restrictions of polynomials.*

A. Foundations and permitted imports

Use quotient rings, the vanishing ideal $I(X)$ of the actual point set, and $V(I(X)) = X$ for an algebraic set: if $X = V(S)$, then $S \subset I(X)$ gives $V(I(X)) \subset X$, while the reverse inclusion is definitional. Constants are fixed by a k-algebra map. No Nullstellensatz or algebraic closure is necessary for this formulation.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-3: N1/E1/v1 to N3. $F^* : k[Y] \to k[X]$ is a well-defined unital k-algebra homomorphism.

Receiving activity. N3.1

Compatibility. E1/v1 and E2/v1; maps $X \xrightarrow{F} Y \xrightarrow{G} Z$. Check: The arrow directions reverse: $(G \circ F)^* = F^* \circ G^*$.

T2-3: N2/E2/v1 to N3. There is a unique polynomial map $F : X \to Y$ with $F^* = \phi$.

Receiving activity. N3.1

Compatibility. E1/v1 and E2/v1; maps $X \xrightarrow{F} Y \xrightarrow{G} Z$. Check: The arrow directions reverse: $(G \circ F)^* = F^* \circ G^*$.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Define the pullback

Object and role. Compose representative polynomials and show the result is independent of representatives. **Local obligation:** $F^* : k[Y] \to k[X]$ is a well-defined unital k-algebra homomorphism.

Inputs. A polynomial map $F = (F_1, \ldots, F_m) : X \to Y$.

C — Contract

Publish E1/v1 only after establishing this obligation: $F^* : k[Y] \to k[X]$ is a well-defined unital k-algebra homomorphism.

K — Knowledge

A polynomial map $F = (F_1, \ldots, F_m) : X \to Y$. Retain the stated assumptions and import versions.

S — Strategy

Compose representative polynomials and show the result is independent of representatives.

V — Validation

Two representatives equal on Y must remain equal after composition because $F(X) \subset Y$.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 For a polynomial h in the Y coordinates, substitute the polynomial coordinates of F to obtain a polynomial function on X. If $h - h' \in I(Y)$, then $(h - h')(F(x)) = 0$ for every $x \in X$, so the resulting classes in $k[X]$ agree.

N1.2 Composition preserves addition, multiplication, the constant 1, and each scalar in k. It also does not depend on the representatives chosen for F_i, because those representatives agree pointwise on X. This proves the pullback assertion.

Export E1/v1. $F^* : k[Y] \to k[X]$ is a well-defined unital k-algebra homomorphism.

Acceptance gate. Two representatives equal on Y must remain equal after composition because $F(X) \subset Y$.

Targeted repair. If composition is defined before target membership is checked, verify $F(X) \subset Y$; otherwise relations in $I(Y)$ need not vanish. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Reconstruct a map from coordinates

Object and role. Choose representatives for the images of coordinate classes and verify every defining relation of Y. **Local obligation:** There is a unique polynomial map $F : X \to Y$ with $F^* = \phi$.

Inputs. A unital k-algebra homomorphism $\phi : k[Y] \to k[X]$.

C — Contract

Publish E2/v1 only after establishing this obligation: There is a unique polynomial map $F : X \to Y$ with $F^* = \phi$.

K — Knowledge

A unital k-algebra homomorphism $\phi : k[Y] \to k[X]$. Retain the stated assumptions and import versions.

S — Strategy

Choose representatives for the images of coordinate classes and verify every defining relation of Y.

V — Validation

The images are elements of the quotient $k[X]$, so changing representatives must not change the point map.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Let $\bar{y}_j$ be the coordinate classes of $k[Y]$. Choose polynomials p_j representing $\phi(\bar{y}_j)$, and set $F(x) = (p_1(x), \ldots, p_m(x))$. Representatives differing by $I(X)$ give the same map.

N2.2 For every $h \in I(Y)$, $h(\bar{y}_1, \ldots, \bar{y}_m) = 0$ in $k[Y]$. Applying ϕ gives $h(p_1, \ldots, p_m) \in I(X)$, hence $F(x) \in V(I(Y)) = Y$. On all coordinate classes F^* agrees with ϕ, and these classes and constants generate $k[Y]$, so agreement holds everywhere. The coordinate images also force uniqueness.

Export E2/v1. There is a unique polynomial map $F : X \to Y$ with $F^* = \phi$.

Acceptance gate. The images are elements of the quotient $k[X]$, so changing representatives must not change the point map.

Targeted repair. If arbitrary polynomial representatives are used without checking relations, apply ϕ to every relation in $I(Y)$ before claiming target membership. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Reverse composition and detect isomorphisms

Object and role. Compute compositions on functions and use uniqueness of the recovered map. **Local obligation:** Pullback gives the stated bijection, reverses composition, and identifies polynomial isomorphisms with algebra isomorphisms.

Inputs. E1/v1 and E2/v1; maps $X \xrightarrow{F} Y \xrightarrow{G} Z$. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: Pullback gives the stated bijection, reverses composition, and identifies polynomial isomorphisms with algebra isomorphisms.

K — Knowledge

E1/v1 and E2/v1; maps $X \xrightarrow{F} Y \xrightarrow{G} Z$. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Compute compositions on functions and use uniqueness of the recovered map.

V — Validation

The arrow directions reverse: $(G \circ F)^* = F^* \circ G^*$.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 For a function h on Z, $(G \circ F)^* h = h \circ G \circ F = F^*(G^* h)$. Identity maps pull back to identity homomorphisms. E1 and E2 are inverse constructions, giving the bijection.

N3.2 If F has polynomial inverse G, the composition identities show that F^* and G^* are inverse. Conversely if F^* is invertible, E2 realizes its inverse as G^* for a polynomial map $G : Y \to X$. Both composite pullbacks are identities; uniqueness in E2 forces $G \circ F = \mathrm{id}_X$ and $F \circ G = \mathrm{id}_Y$.

Export E3/v1. Pullback gives the stated bijection, reverses composition, and identifies polynomial isomorphisms with algebra isomorphisms.

Acceptance gate. The arrow directions reverse: $(G \circ F)^* = F^* \circ G^*$.

Targeted repair. If composition order is reversed incorrectly, evaluate both sides on one function and rebuild the domain/codomain chain. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Define the pullback

1. **N1.1.** For a polynomial h in the Y coordinates, substitute the polynomial coordinates of F to obtain a polynomial function on X. If $h - h' \in I(Y)$, then $(h - h')(F(x)) = 0$ for every $x \in X$, so the resulting classes in $k[X]$ agree.

2. **N1.2.** Composition preserves addition, multiplication, the constant 1, and each scalar in k. It also does not depend on the representatives chosen for F_i, because those representatives agree pointwise on X. This proves the pullback assertion.

B. N2 — Reconstruct a map from coordinates

1. **N2.1.** Let $\bar{y}_j$ be the coordinate classes of $k[Y]$. Choose polynomials p_j representing $\phi(\bar{y}_j)$, and set $F(x) = (p_1(x), \ldots, p_m(x))$. Representatives differing by $I(X)$ give the same map.

2. **N2.2.** For every $h \in I(Y)$, $h(\bar{y}_1, \ldots, \bar{y}_m) = 0$ in $k[Y]$. Applying ϕ gives $h(p_1, \ldots, p_m) \in I(X)$, hence $F(x) \in V(I(Y)) = Y$. On all coordinate classes F^* agrees with ϕ, and these classes and constants generate $k[Y]$, so agreement holds everywhere. The coordinate images also force uniqueness.

C. N3 — Reverse composition and detect isomorphisms

1. **N3.1.** For a function h on Z, $(G \circ F)^* h = h \circ G \circ F = F^*(G^* h)$. Identity maps pull back to identity homomorphisms. E1 and E2 are inverse constructions, giving the bijection.

2. **N3.2.** If F has polynomial inverse G, the composition identities show that F^* and G^* are inverse. Conversely if F^* is invertible, E2 realizes its inverse as G^* for a polynomial map $G : Y \to X$. Both composite pullbacks are identities; uniqueness in E2 forces $G \circ F = \mathrm{id}_X$ and $F \circ G = \mathrm{id}_Y$.

□

V. WORKED ORCHESTRATION

Over $k = \mathbb{C}$, for the parabola $Y = V(y - x^2)$, the map $F : \mathbb{A}^1 \to Y$, $t \mapsto (t, t^2)$, induces $F^* : k[x, y]/(y - x^2) \to k[t]$ with $x \mapsto t$, $y \mapsto t^2$. Its inverse sends t to the class of x. The corresponding geometric inverse is $(x, y) \mapsto x$. Both compositions are explicitly identities.

A. Boundary and hypothesis audit

A pointwise bijective polynomial map need not be an isomorphism. Over an algebraically closed field of characteristic $p > 0$, $t \mapsto t^p$ is bijective: roots exist and $(a - b)^p = 0$ forces equality. Its pullback $k[t] \to k[t]$, $t \mapsto t^p$, is not surjective because t is absent from $k[t^p]$.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected final export: every bijective polynomial map has a polynomial inverse. **Validation witness:** Frobenius on the affine line in characteristic p. **Diagnosis:** Contract promoted a set-theoretic property to an algebraic one. **Repair:** use the actual criterion that the pullback is an algebra isomorphism, as proved in N3. This rejects a stronger claim while preserving the exact §1 theorem; bijectivity alone remains insufficient.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Extend this affine viewpoint to gluing local coordinate descriptions, while keeping regular maps and pointwise bijections distinct.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. J. S. Milne, Algebraic Geometry, chapters on commutative algebra, algebraic sets, and affine varieties; author's mathematics site.